

CRUISE AND FERRY TERMINALS

Guaranteed Passage

Cruise and ferry terminals as real estate and infrastructure,
and a test of when a sea gate earns its land in Egypt, Saudi
Arabia and the Gulf



SIX NUMBERS THAT FRAME THE ARGUMENT

22:1

ferry passengers for every cruise passenger in EU ports in 2024: 417.8 million sea passengers in all, 17.9 million of them on cruises

3.0x

what a homeport cruise passenger spends ashore in Europe compared with a transit passenger: €212.67 against €70.63

US\$2.5bn

Royal Caribbean's 36-year terminal usage agreement at PortMiami, with 2.1 million passenger moves a year guaranteed from year 11

6%

share of Galataport's targeted 25 million annual visitors who would be cruise passengers or crew

962

average passengers per cruise call in Egypt in 2023, against 2,236 across MedCruise member ports in 2025

1.5%

annual growth implied by CLIA's outlook of 40 million cruise passengers by 2030, after a record 37.2 million in 2025

CONTENTS

Summary	3
1 Two trades, one quay	4
2 Where the cruise begins	7
3 The guarantee is the asset	10
4 What the terminal anchors	13
5 Designing in the limits	16
6 The Heirstone Sea Gate Test	18
7 Egypt, Saudi Arabia and the Gulf	22
Recommendations	25
Notes, sources and definitions	26

ABOUT THIS REPORT

Heirstone Consulting advises developers, sovereign and institutional investors, port and land authorities and tourism bodies across the Middle East and Africa. This report uses only public sources current at October 2026: port authority and operator releases, government statistics, concession announcements, listed-company filings, industry statistics and press reports that quote the parties involved.

Where we combine published figures to produce a new one, such as passengers per call, port revenue per passenger move or capital per annual passenger, the calculation is labelled Heirstone analysis and its inputs are shown. Figures reported by a single trade or press source are attributed in the text.

The Heirstone Sea Gate Benchmark and the Sea Gate Test in chapter 6 are our own. They score gates only on what is publicly documented.

heirstoneconsulting.com
info@heirstoneconsulting.com

FIVE TYPES OF SEA GATE, FROM THE HEIRSTONE SEA GATE TEST (CHAPTER 6)

Underwritten gate

Traffic is committed and passengers start or end trips here. Build to the contract.

PortMiami, Helsinki

Owned stop

A line commits transit calls to a stop it controls. Let it build and pay rent or equity.

Celebration Key

Open homeport

Nothing committed, but a third or more of passengers turn round here. Sell turnarounds as contracts.

Piraeus, Doha

Anchor gate

Transit traffic with land behind the quay. Underwrite the property.

Galataport, Valletta

Call stop

Transit traffic and no land in play. Keep capital light and levy each visitor.

Dubrovnik

SUMMARY

Sea gates pay their way where someone has committed the traffic before the concrete is poured

Port authorities in the Red Sea and the Gulf describe new terminals by annual capacity. The ports that have made terminals pay describe them by what is committed: passenger moves guaranteed by a cruise line, a terminal a line has paid for, or a ferry timetable that runs all year. Where nothing is committed, a sea gate's value lies in the land behind it, and the terminal is a public cost to be kept light and priced.

1 EU ports see about 22 ferry passengers for every cruise passenger.

EU ports handled 417.8 million sea passengers in 2024, of whom 17.9 million were on cruises. Messina's strait ferries carried 11.4 million people in 2024, almost three times Barcelona's 3.99 million cruise passengers in 2025. Ferries need vehicle lanes and frequency; cruise needs hotel beds, flights and a hall that can clear a large ship in a morning.

2 Turnaround passengers spend three times as much ashore.

In Europe a homeport passenger spends €212.67 ashore, a transit passenger €70.63. At those rates a turnaround call by an Icon-class ship is worth about €1.2 million ashore and an average Egyptian call about €68,000. Barcelona and Piraeus handle 58 to 59% of their traffic as homeport movements; the MedCruise average is 27%.

3 The busiest US terminals are underwritten by the lines.

Royal Caribbean committed US\$2.5 billion over 36 years for PortMiami's Terminal G, with 2.1 million passenger moves a year guaranteed from year 11. MSC built its US\$300 million terminal itself. Galveston financed its newest terminal against line contracts. On our analysis, Miami's line agreements pay the port about US\$14 to US\$36 per passenger move.

4 Without a guarantee, the property has to carry the case.

Galataport's own targets put cruise passengers and crew at 6% of the 25 million visitors it expected each year, so the other 94% carry its business case. Helsinki moved its containers out in 2008, built a district of more than 100 hectares beside its ferry harbour and gave its new terminal 37,000 m² of building rights.

5 European ports are paying to retrofit limits.

Barcelona is spending €185 million on a programme that cuts its daily cruise capacity by 16%; Venice budgeted €157 million for temporary berths after its 2021 ban. Levies of €5 to €20 a head are now common in Europe. We found no published per-passenger cruise levy in Egypt, Saudi Arabia or the UAE.

6 The region is adding capacity faster than commitments.

Three Egyptian Red Sea terminals concessioned in 2024 have a stated capacity of 4.3 million passengers a year; Egypt's six cruise ports handled 198,124 in 2023. Cruise Saudi targets 1.3 million cruise passengers a year by 2035, about nine times the 140,000 that its line AROYA carried in its first year, according to Cruise Arabia & Africa. Doha shows the alternative: 56 of its 87 calls in 2024/25 were full or partial turnarounds.

THE HEIRSTONE SEA GATE TEST IN BRIEF (CHAPTER 6)

1

Committed traffic

A line guarantee or line funding, or a ferry timetable carrying a million a year

2

Turnaround share

A third or more of passengers start or end their trip at the gate

3

Capture per move

A published tariff, rent or levy worth about US\$10 or more a move

4

Land in play

Developable land or building rights beside the terminal, in one plan

5

Load limits set

Daily caps and a levy published, ideally before the first call



CHAPTER 1

01

Two trades, one quay

Ferries and cruise ships often share a harbour. They differ in almost everything a planner has to size: frequency, vehicles, time in port and spending ashore.

22:1

ferry to cruise passengers in EU ports, 2024 (Heirstone analysis of Eurostat data)

EU ports handled 417.8 million sea passengers in 2024. Cruise passengers accounted for 17.9 million of them, so for every cruise passenger roughly 22 people travelled by ferry. The busiest passenger port in the EU is Messina, where the strait ferries carried 11.4 million people, almost three times the 3.99 million cruise passengers who passed through Barcelona, the Mediterranean's busiest cruise port, in 2025.

Sixty-four per cent of EU ferry traffic runs between ports of the same country, on island lines and strait crossings. These are the high-volume gates. They run all year,

carry vehicles and freight with their passengers, and are sized by lanes, sailings an hour and peak-day queues.

Cruise is the smaller trade by volume and the larger by spending per head. Its passengers arrive in thousands, a few days a week, and what they spend depends on whether their trip begins at the port or merely pauses there.

Planning a sea gate starts with knowing which trade is coming and what each will ask of the land around the quay. This chapter sets out the volumes, the cruise industry's own outlook and the ships a new terminal must now receive.

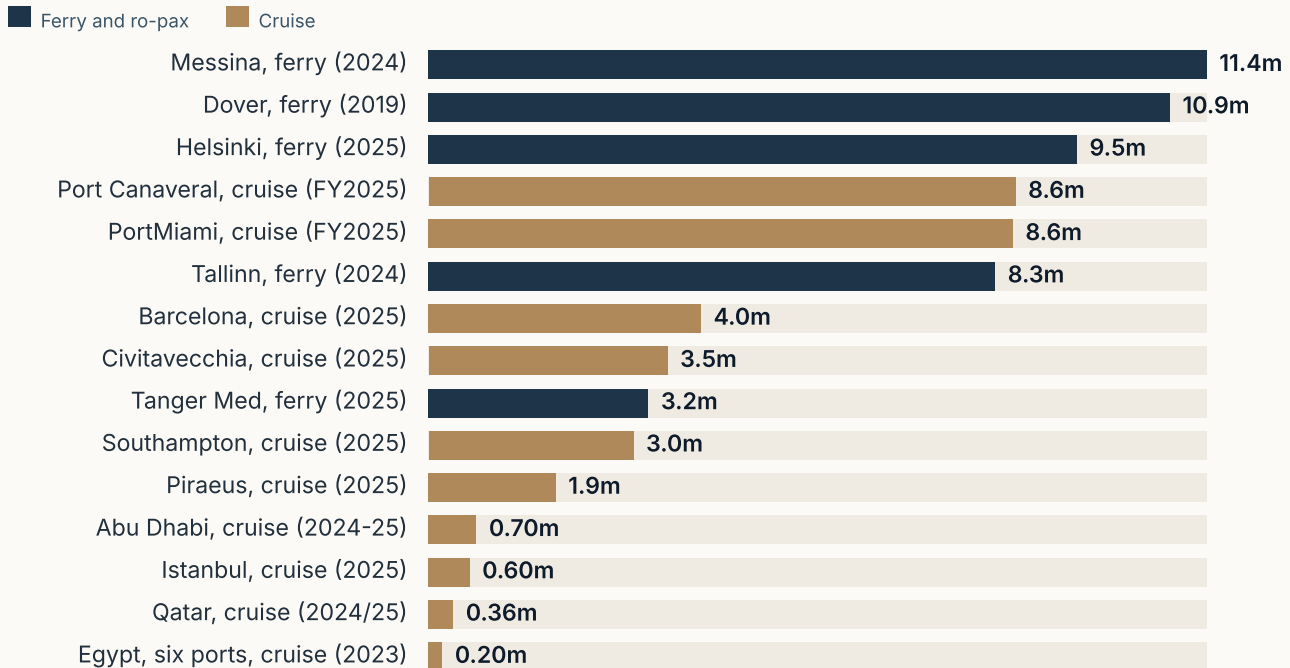
VOLUME

Europe's busiest ferry ports each handle more than twice the passengers of its busiest cruise port

EXHIBIT 1

Ferry gateways out-carry all but the two largest cruise homeports

Passengers through the port, millions, latest published year (ferry ports: embarked and disembarked; cruise: passenger movements or visitors as published)



Source: Eurostat, Maritime passenger statistics (2024 data); Wikipedia, Port of Dover (Dover Harbour Board) and Port of Tallinn; Port of Helsinki (Jan 2026); Tanger Med Port Authority (2025); Cruise & Ferry (Dec 2025) and Royal Caribbean Blog quoting Miami-Dade County and Canaveral Port Authority (Dec 2025); Risposte Turismo via Cruise & Ferry (Apr 2026); ABP via Ports Europe (Southampton, more than 3 million, 2026); Greek Reporter on MedCruise (Jul 2026); CruiseMapper citing DCT Abu Dhabi (Jan 2026); Cruise Arabia & Africa (May 2025); Seatrade Cruise News quoting Egyptian Ports (Nov 2024). Bases differ between sources.

Helsinki shows the gap at a single port. In 2025 it handled 9.5 million ferry passengers, 7.5 million of them on the shuttle to Tallinn, and 175,000 international cruise guests on 101 calls, about 1.8% of its 9.6 million passengers. It also handled 1.4 million passenger cars and 462,000 lorries. Dover handled 10.9 million passengers in 2019 and, in 2017, about 2.6 million lorries and 2.2 million cars.

The Strait of Gibraltar adds a seasonal peak. Spain's summer operation for travellers to North Africa counted 3,488,208 passengers and 857,851 vehicles between 15 June and 15 September

2025. Morocco counted 3.2 million passengers and 739,000 vehicles through its ports between 10 June and 15 September, 1.8 million of them at Tanger Med.

Urban water transport is a niche even in water cities. Dubai's abras, water buses, water taxis and ferry carried 18.4 million riders in 2025, about 2.3% of the 802 million trips on the emirate's public transport, shared mobility and taxis. Istanbul's municipal ferry company says it carries more than 40 million passengers a year; in 2021 its management put sea transport at about 3% of the city's public transport trips.

79%

of Helsinki's ferry passengers used the Tallinn route in 2025 (7.5m of 9.5m)

1.8%

of all passengers through the Port of Helsinki in 2025 were cruise guests

2.3%

of Dubai's 802 million public transport and taxi trips in 2025 were by abra, water bus, water taxi or ferry

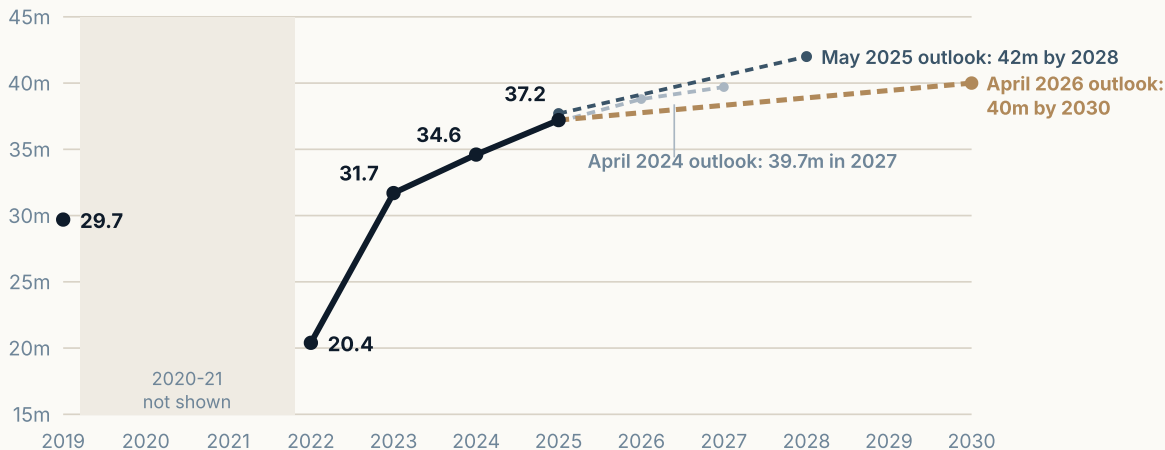
CRUISE DEMAND AND SHIP SIZE

Cruise passenger numbers set a record in 2025 while the industry's own outlook flattened

EXHIBIT 2

Record years, flatter outlook: CLIA's latest forecast implies about 1.5% a year after 2025

Global ocean cruise passengers, millions: actual, and forecasts by the CLIA report that published them

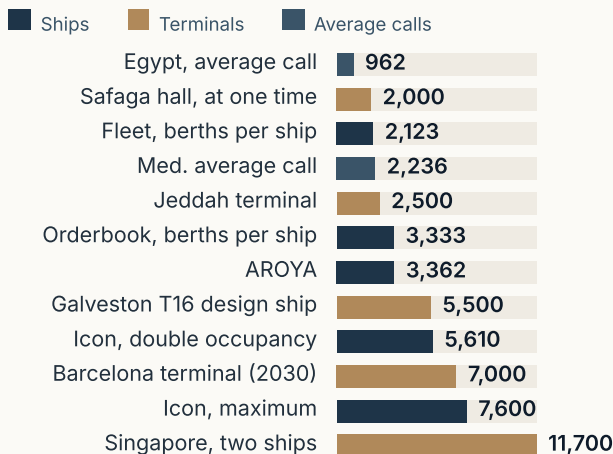


Source: CLIA, State of the Cruise Industry reports (April 2024, May 2025) and press release (15 April 2026); TravelPulse and Seatrade Cruise News on the 2025 and 2026 reports. Forecast lines join published points only. Heirstone analysis: $(40 / 37.2)^{1/5} - 1 = 1.5\%$ a year; $(37.2 / 29.7)^{1/6} - 1 = 3.8\%$ a year for 2019 to 2025.

EXHIBIT 3

New ships are larger than several of the region's halls

Passengers per ship (CLIA fleet 2026; orderbook to 2037), per average call (Egypt 2023; Mediterranean 2025) or per terminal at one time



Source: CLIA (Apr 2026; orderbook via eGlobal Travel Media, Jun 2026); Wikipedia (Icon of the Seas); Gulf Business (AROYA, more than 3,362); Seatrade Cruise News (Egypt, MedCruise); The Maritime Executive (Safaga, Jan 2024; Galveston, Nov 2025); Ship Technology (Jeddah, more than 2,500, Jul 2021); Port de Barcelona (Jul 2025); Global Ports Holding (Singapore). Heirstone analysis: 690,000 berths / 325 ships; 200,000 / 60 ships.

CLIA counted 37.2 million ocean cruise passengers in 2025, 7.5% more than in 2024 and well above the 29.7 million of 2019. Its outlook has moved the other way. The 2025 report forecast 42 million passengers by 2028; the 2026 report forecasts 40 million by 2030, which implies growth of about 1.5% a year from 2025. The 2025 report had also forecast 37.7 million for 2025 itself, half a million more than were carried.

Terminal business cases written on high single-digit growth are therefore exposed. The fleet will keep growing, mostly through larger ships. CLIA members operate 325 ocean ships with about 690,000 lower berths in 2026, about 2,100 per ship. The 60 ships on order to 2037 add nearly 200,000 berths, about 3,300 each, at a cost of about US\$71 billion, or US\$355,000 per berth.

The reference ship for a new terminal is now Icon of the Seas: 365 metres long, 5,610 guests at double occupancy and up to 7,600 in all. A line will send a ship that cost €1.86 billion only to a port that can clear its passengers and luggage in a morning. Two of the region's halls for which ratings are published, Safaga and Jeddah, are rated at 2,000 and more than 2,500 passengers, fewer than AROYA carries.

CHAPTER 2

02

Where the cruise
begins

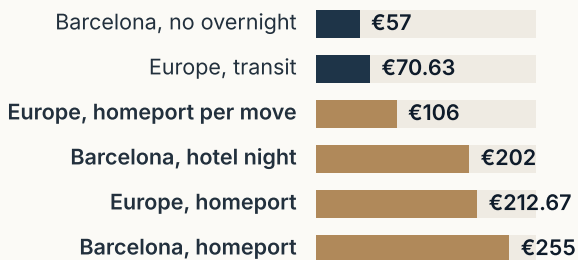
A passenger who starts or ends a cruise in a city spends about three times as much there as one who calls for the day.

Ports report passengers. Cities earn from the hotel nights, flights, tours and meals that come with them, and most of those follow the passengers whose trip begins or ends at the port. For a sea gate, turnaround share is the number to underwrite.

EXHIBIT 4

Homeport passengers spend about three times
as much ashore

Spend in the port region per passenger or per movement, euros: Europe 2019 basis; Barcelona 2014 study and 2024-26 estimate



Source: Cruise Europe fact sheet on CLIA/BREA economic impact studies (220 studies, 2019 basis); Port de Barcelona and Turisme de Barcelona (2014) via PortEconomics; Euronews Travel (Jun 2026). **Heirstone analysis:** per homeport movement = €212.67 / 2.

The broadest European evidence comes from 220 economic impact studies summarised by Cruise Europe on a 2019 basis. A homeport passenger spends €212.67 in the port region, a transit passenger €70.63 and a crew member €26.49. The homeport figure includes hotel nights: 58% of homeport passengers stay in the destination before or after the cruise, for an average of 2.3 nights.

Barcelona shows the same gap. A 2014 study for the port and the city's tourism board found that cruise passengers with a hotel night spent €202 and those without €57. A University of Barcelona study for the port and CLIA found that US homeport passengers, 28.5% of homeport passengers in 2024, spent about €307 a day over 2.8 nights. Cruise passengers paid nearly €261 million in taxes that year.

A homeport passenger on a round trip embarks and disembarks at the same port and so appears twice in its statistics. Spend per movement is therefore about €106, still half as much again as a transit passenger. That is the like-for-like figure to use when ports are compared by passengers.

The University of Barcelona study counted 2.8 million unique cruise passengers in Barcelona in 2024 and 3.7 million movements, so each passenger passed through the port 1.3 times on average. A port that reports movements will look larger than one that reports visitors, and the gap widens with its homeport share. Comparisons in this report state the basis each port uses.

69%

of cruisers stayed at least one room-night before or after their cruise (CLIA 2024 survey); 60% before and 54% after

1.32

port movements per unique cruise passenger in Barcelona in 2024: 3.7 million movements by 2.8 million people (Heirstone analysis)

€1.2bn

generated in Catalonia by cruise activity in 2024, with €707m of GDP and more than 9,500 jobs (University of Barcelona)

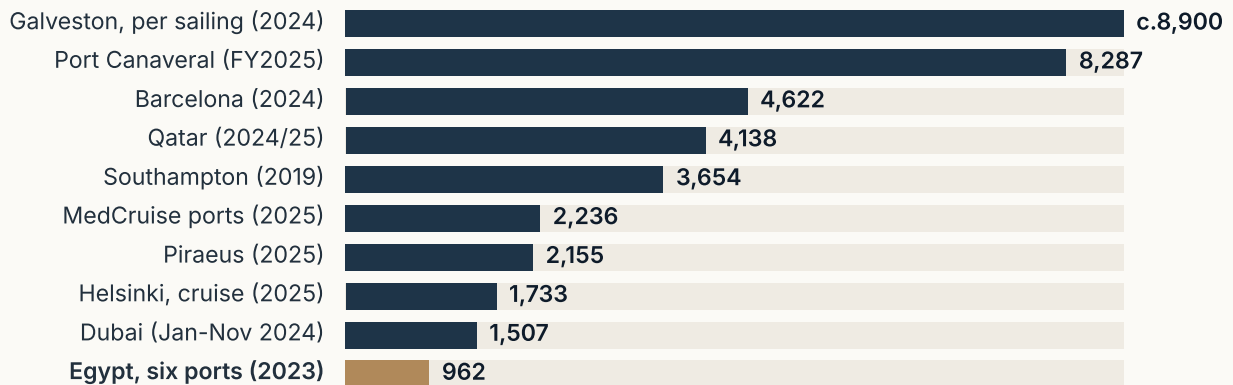
VALUE PER CALL

A single turnaround call by an Icon-class ship is worth about as much ashore as eighteen average Egyptian calls

EXHIBIT 5

Homeport calls in Florida and Texas put about nine times as many passengers through the port as an average Egyptian call

Published passengers or movements divided by calls or sailings, latest year

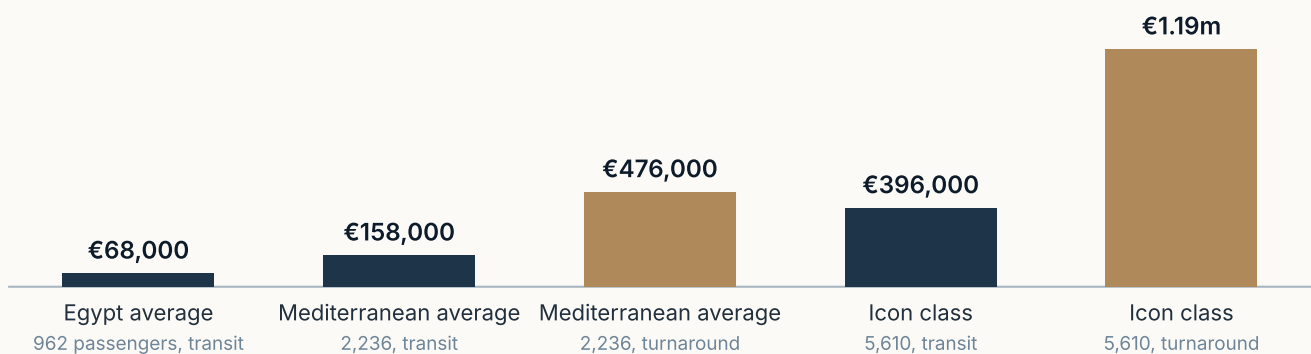


Source: Canaveral Port Authority (Nov 2025); Houston First (2025); Seatrade Cruise News on MedCruise (2025, 2026); Cruise Arabia & Africa quoting Mwani Qatar and Qatar Tourism (May 2025); Wikipedia, Port of Southampton; Greek Reporter (Jul 2026); Port of Helsinki (Jan 2026); Dubai Media Office (Dubai Customs, Dec 2024); Seatrade Cruise News quoting Egyptian Ports (Nov 2024). **Heirstone analysis.** Homeport ports count each passenger twice (embarking and disembarking); Dubai counts customs-processed passengers; Galveston is more than 3.4m on more than 380 sailings.

EXHIBIT 6

At European spend rates, one turnaround call is worth three transit calls of the same ship

Illustrative spending ashore per call, euros: passengers per call multiplied by European average spend per passenger



Source: Heirstone analysis using Cruise Europe/BREA spend rates (2019 basis: transit €70.63, homeport €212.67), Egyptian Ports (962 passengers per call, 2023), MedCruise (2,236 per call, 2025) and Icon of the Seas (5,610 at double occupancy). A turnaround call in steady state adds the full homeport spend of one shipload. We found no published spend survey for any Gulf, Red Sea or Egyptian port.

Exhibit 5 divides each port's published passengers by its calls. The Florida and Texas homeports put roughly 8,300 to 8,900 passengers through the port on each call, because most passengers end or begin their cruise there. Barcelona averaged about 4,600 movements per call in 2024 and the MedCruise average was 2,236 in 2025. Egypt averaged 962 in 2023, consistent with smaller ships and few turnarounds.

Exhibit 6 turns those passengers into spending. The rates are European and the result is illustrative, but the order of magnitude holds: an average Egyptian call is worth about €68,000 to the port region and an Icon-class turnaround about €1.2 million. Most of that money is spent outside the terminal: in hotels, on tours, in restaurants and on flights.

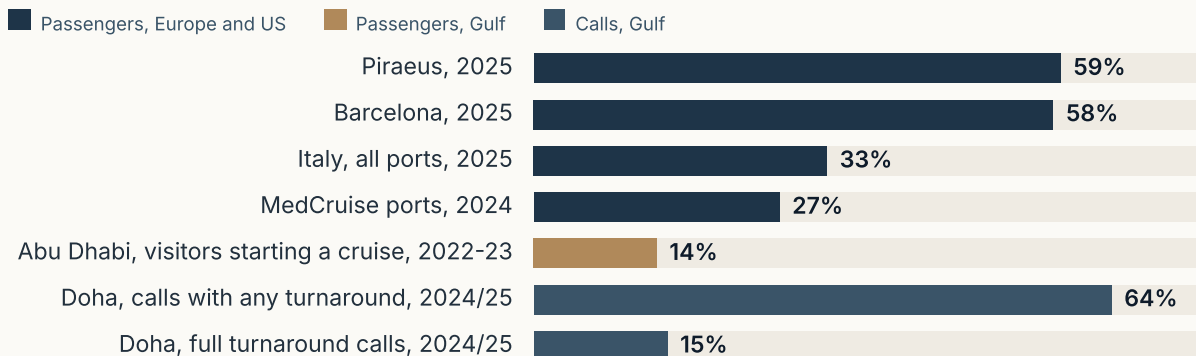
HOMEPORT SHARE

The Mediterranean's leading cruise gates are majority homeport, and Gulf ports already run turnarounds

EXHIBIT 7

Homeport movements are 58 to 59% of traffic at Barcelona and Piraeus; 14% of Abu Dhabi's visitors started their cruise there

Homeport movements as a share of all cruise passenger movements, and turnaround calls as a share of calls



Source: Greek Reporter on MedCruise (Jul 2026); Seatrade Cruise News on MedCruise (May 2025); Risposte Turismo via Cruise & Ferry (Apr 2026); Alethiad/WAM (Nov 2023); Cruise Arabia & Africa quoting Mwani Qatar and Qatar Tourism (May 2025). **Heirstone analysis.** Barcelona combines MedCruise homeport movements (2,311,722) with the Risposte Turismo total (3.99m).

Barcelona handled 2.31 million homeport movements in 2025 and Piraeus 1.09 million, 58 to 59% of their traffic. Across MedCruise ports, homeport movements were 27% of the total in 2024; in Italy they were 33% in 2025. The Barcelona studies tie its homeport role to hotel nights and air links: its US homeport passengers stayed 2.8 nights on average.

The Gulf is already a turnaround market. In Qatar's 2024/25 season, 13 of 87 calls were full turnarounds and 43 partial ones, according to Mwani Qatar and Qatar Tourism figures reported by Cruise Arabia & Africa; the season drew more than 360,000 cruise visitors. In Abu Dhabi, more than 82,000 of the

600,000 or more cruise visitors of 2022-23 started their cruise there. The emirate's upside lies in turning more of its visitors into embarkations, which depends on flights and hotel packages as much as on berths.

For a transit port, the money that reaches the destination goes mostly to what is within reach of the pier. In the Caribbean and Latin American study for the cruise association FCCA, shore excursions were the largest item, at US\$40.44 of an average US\$104.36 per passenger visit, ahead of watches and jewellery at US\$24.44 and food and drink at US\$9.72. The published results do not separate spending inside terminals.

39%

of a Caribbean cruise visitor's US\$104.36 ashore goes on shore excursions (2023/24)

83%

of transit passengers went ashore in the 2024 Caribbean study, down from 85% in 2018

5.7 hrs

average time ashore for a transit cruise visitor in Barcelona (Euronews, 2026)

CHAPTER 3

03

The guarantee is the asset

PortMiami's terminals were built on contracts. Cruise lines either paid for them or guaranteed the passengers who would use them, and the county set rents and fees against those commitments.

PortMiami handled a record 8,564,225 cruise passengers in the year to September 2025, and Port Canaveral 8,602,047 passenger movements. PortMiami's ten terminals include dedicated facilities for MSC, Royal Caribbean, Norwegian, Carnival and Virgin Voyages. In 2023-24 Royal Caribbean and Miami-Dade County agreed a 50-year package worth US\$2.8 billion: US\$2.5 billion from a cruise terminal usage agreement for the new Terminal G and US\$260 million from an amended lease on the line's headquarters campus.

The terminal agreement is a volume contract. Royal Caribbean guarantees 600,000 passenger moves a year while Terminal G is built, 1.5 million on completion and 2.1 million a year from year 11 to the end of the 36-year term. The port builds the terminal for the line's exclusive use.

That guarantee is what makes the terminal financeable. It turns a building into a stream of payments from a known counterparty, which a port can borrow against and a lender can price.

According to IAPH's WorldPorts, the passenger moves guaranteed at PortMiami will exceed the port's record 2019 financial-year volume by 22%.

We found no published agreement of this kind for a cruise terminal in Egypt, Saudi Arabia or the UAE. The rest of this chapter sets out the forms these contracts take, what each passenger move pays the port, and how long the terms run.

WHO PAYS

In Miami, Galveston and Southampton, lines pay for terminals or guarantee the passengers who will use them

EXHIBIT 8

Who pays for the terminal, for how long, and what is committed

Selected cruise terminal agreements and concessions with published terms

GATE AND TERMINAL	LINE OR OPERATOR	OPENED OR SIGNED	CAPEX	WHO PAYS	TERM	TRAFFIC COMMITTED	PAYMENT TO THE PORT
CRUISE LINE UNDERWRITES THE TERMINAL							
PortMiami, Terminal G	Royal Caribbean	c.FY2027	n/p	Port builds for line's exclusive use	36 yrs (50-yr package)	0.6m moves a year in build, 1.5m on opening, 2.1m from year 11	US\$2.5bn usage; US\$260m HQ lease
PortMiami, Terminal AA	MSC	Apr 2025	c.US\$300m	MSC	50 or 62 yrs	n/p	c.US\$1.7bn ground rent; US\$6.19 a move above 2.5m
PortMiami, Terminal B	Norwegian	n/p	US\$215-235m	County fronts; line repays in 25 yrs	Lease to 2050	1.3m; 1.5m by 2024; 1.8m by 2031	c.US\$24m a year (FY2018), +3% a year
PortMiami, Terminal V	Virgin Voyages	n/p	to US\$150m	n/p	n/p	n/p	c.US\$12.5m a year (2022), +3% a year
Galveston, Terminal 16	MSC, NCLH	Nov 2025	US\$156m	Port, on line contracts	NCL 10 yrs + options	Long-term contracts	n/p
Southampton, Horizon	MSC, NCLH	Jul 2021	£55m	ABP with the lines	n/p	n/p	n/p
PORT OR CONCESSIONAIRE CARRIES THE TRAFFIC RISK							
Port Canaveral	Several	FY2026-30	US\$912m plan	Port authority	n/a	18 homeported ships planned	US\$182m cruise revenue (FY2025)
Barcelona, Adossat	Several	2026-30	€185m	Public and private	n/p	Capacity cut to 31,000 a day	n/p
Istanbul, Galataport	Doğuş (operator)	Oct 2021	US\$1.7bn incl. bid	Concessionaire	30 yrs	None published	US\$702m bid (2013)
Safaga, Hurghada, Sharm	AD Ports (operator)	Jun 2024	US\$3m-4.7m for three	Concessionaire	15 yrs	None published; 4.3m stated capacity	n/p
Ain Sokhna, ro-ro and cruise	AD Ports (operator)	Jun 2024 (initialled)	US\$5.3m reported	Concessionaire	30 yrs	n/p	n/p

Source: WorldPorts (IAPH); Miami Today; Travel Agent Central (2018); Wikipedia, PortMiami (MSC term 62 years; Miami Today gives 50); The Maritime Executive (Nov 2025); Baird Maritime and Cruise Britain (2021); Canaveral Port Authority (Nov 2025); Port de Barcelona (Jul 2025); Hürriyet Daily News (May 2013); Galataport Istanbul; Global Ports Holding FY2024 results; AD Ports Group and Abu Dhabi Media Office (Jun 2024). A 2018 report gave US\$100m of county investment in Terminal B. Egypt: US\$4.7m at signing (Jun 2024); US\$3m in The Maritime Executive (Jan 2024). n/p: not published.

The agreements take three forms. In the first, the line builds and pays: MSC built Terminal AA at PortMiami itself, for about US\$300 million on 16.7 acres with two berths, and pays about US\$1.7 billion of ground rent over 50 years plus US\$6.19 for each passenger move above 2.5 million a year.

In the second, the port builds and the line repays or guarantees. Miami-Dade fronted the cost of Norwegian's Terminal B, which Norwegian repays over 25 years, and Norwegian guaranteed 1.3 million passenger moves a year, rising to 1.8 million by 2031.

In the third, a public port borrows on the strength of line contracts. Galveston's US\$156 million Terminal 16, opened in November 2025, was funded on the back of long-term agreements with MSC and Norwegian Cruise Line Holdings. Southampton's £55 million Horizon terminal was built by Associated British Ports in partnership with the same two groups.

Outside the US, many terminals are run by concessionaires with no line commitment. Global Ports Holding handled 13.4 million passengers across 28 ports in the year to March 2024, on calls it does not control.

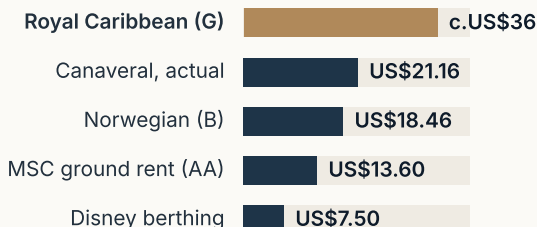
PRICING THE COMMITMENT

Miami's line agreements pay the port about US\$14 to US\$36 for each passenger move

EXHIBIT 9

Royal Caribbean's Terminal G agreement pays the port the most per passenger move

US\$ paid to the port per passenger move, by agreement or port



Source: Heirstone analysis of WorldPorts (IAPH), Miami Today, Travel Agent Central and Canaveral Port Authority. Royal Caribbean: US\$2.5bn / at most 69.6m guaranteed moves over 36 years (10 at up to 1.5m, 26 at 2.1m). Canaveral: US\$182m cruise revenue / 8.6m movements (FY2025). Norwegian: US\$24m / 1.3m (FY2018). MSC: US\$1.7bn / 50 years / 2.5m moves, before wharfage. Disney: US\$2.7m / 360,000 passengers projected for 2023. Ignores ramp-ups, escalation and discounting.

Royal Caribbean's usage payments total US\$2.5 billion over 36 years, about US\$69 million a year. Its guarantees add up to at most 69.6 million moves over the term, so the port receives at least about US\$36 a guaranteed move. Norwegian's minimum guarantee came to about US\$18.50 a move in 2018, rising 3% a year. MSC's ground rent averages US\$34 million a year, about US\$13.60 a move at the 2.5 million threshold, before wharfage. Disney's berthing deal, with no published guarantee, was projected to yield about US\$7.50 per passenger. Port Canaveral's US\$182 million of cruise revenue in its 2025 financial year came to about US\$21 per movement.

The Egyptian Red Sea concessions sit at the other end. In June 2024 AD Ports Group took 15-year concessions to operate the cruise terminals at Safaga, Hurghada and

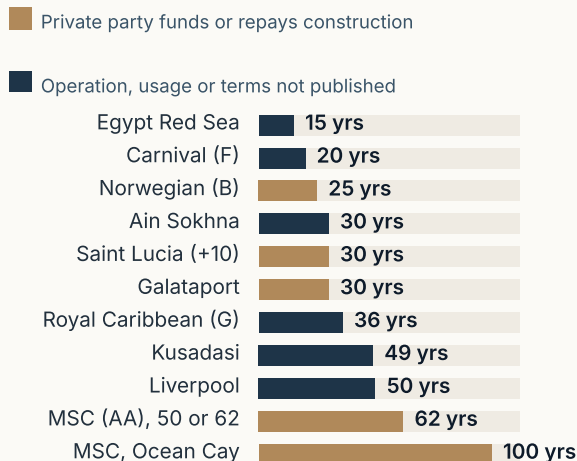
Sharm El Sheikh, with US\$4.7 million of investment over the term according to AD Ports (a January 2024 report gave US\$3 million), or US\$0.2 to 0.3 million a year for three terminals. Royal Caribbean's Miami agreement commits about US\$69 million a year to one. The instruments differ, and the Egyptian terminals already exist, but the comparison shows where the traffic risk sits: in Miami with the line, on the Red Sea with the operator and the port authority.

The Egyptian Red Sea operating concessions run 15 years (Exhibit 10). Every agreement shown under which the private party builds or repays the terminal runs 25 years or more, up to 100 years for MSC's lease of Ocean Cay in the Bahamas, where it planned to spend US\$200 million.

EXHIBIT 10

Where the private side funds the terminal, terms run 25 years or more

Term of selected terminal agreements and concessions, years



Source: AD Ports Group and Abu Dhabi Media Office (Jun 2024); Wikipedia, PortMiami and Ocean Cay; WorldPorts (IAPH); Miami Today; Hürriyet Daily News (2013); Global Ports Holding FY2024 results. MSC (AA): 62 years per Wikipedia, 50 per Miami Today.

US\$69m

average annual usage payment under Royal Caribbean's 36-year Terminal G agreement (Heirstone analysis)

US\$0.3m

average annual investment under the three Egyptian Red Sea concessions (US\$4.7m over 15 years; US\$0.2m on the US\$3m reported in January 2024)

+22%

by which PortMiami's guaranteed passenger moves will exceed its record 2019 financial-year volume (WorldPorts)

CHAPTER 4

What the terminal anchors

Where no line has committed traffic, the land behind the quay has to carry the case.

Galataport is the clearest case of a cruise terminal used to anchor a property scheme.

Doğuş Holding won the 30-year operating right to Istanbul's Salıpazarı port area in 2013 with a bid of US\$702 million, for about 100,000 m² of land and 1.2 km of shore. The operator puts the total project cost at US\$1.7 billion including the bid; other sources give US\$1.8 billion.

The terminal is underground and was designed for 15,000 passengers and crew a day, across three ships. Above it sit 52,000 m² of food and retail in about 250 units, 43,000 m² of offices and a 177-room Peninsula hotel. The operator targeted 25 million visitors a year, of whom 1.5 million would be cruise passengers and crew. Istanbul handled about 596,000 cruise passengers in 2025, up 48% on the year.

On the operator's own targets, the 94% of visitors who are not cruise passengers or crew carry the business case, so they need to be modelled with the same care as the berth. The operating company's debt has since been restructured: bne IntelliNews reported in December 2024 that, after a second restructuring, six Turkish banks took a 49% stake, with about €1.5 billion owed including accrued interest.

Smaller versions of the model work on the same logic. Valletta Cruise Port, majority-owned by Global Ports Holding since 2015, wraps 19 restored 18th-century warehouses and bastions along its quay and receives more than half a million cruise passengers a year. Lisbon's 13,800 m² terminal, opened in 2017 on a 1,490-metre pier, is 500 metres from the city centre. Cape Town's cruise terminal sits beside the V&A Waterfront.

6%

of Galataport's targeted 25 million annual visitors would be cruise passengers or crew

US\$23.4m

a year: the 2013 bid of US\$702m spread over the 30-year right (Heirstone analysis)

15,000

passengers and crew a day: the design capacity of Galataport's underground terminal, across three ships

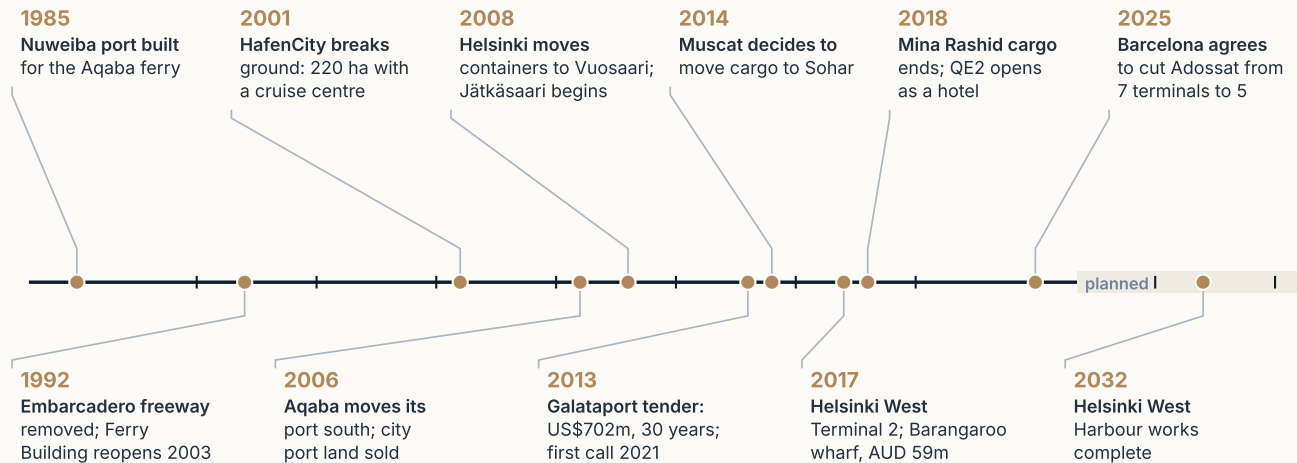
CARGO OUT, PASSENGERS IN

Helsinki moved its containers out first; the district and the new terminal followed

EXHIBIT 11

At Helsinki, Aqaba, Muscat and Mina Rashid, cargo moved out before the waterfront moved in

Selected port land releases, passenger terminals and waterfront schemes, 1985 to 2032



Source: Port of Helsinki development programme; Wikipedia: Jätkäsaari, San Francisco Ferry Building, HafenCity, Port of Aqaba, Port Sultan Qaboos, Mina Rashid, Nuweiba, Barangaroo ferry wharf and Galataport; PES-Architects; Hürriyet Daily News (2013); Port de Barcelona (Jul 2025). Aqaba and Muscat dates are as given in Wikipedia's port histories.

Helsinki's West Harbour shows the sequence for a ferry gate. Container traffic moved to the new Vuosaari harbour in November 2008. The cargo port at Jätkäsaari became a district of more than 100 hectares planned for 17,000 to 18,000 residents and 6,000 jobs, with a 16-storey hotel beside the terminals from 2016. West Terminal 2, with 12,800 m² of floor space, opened in 2017 for Tallinn sailings; the Helsinki-Tallinn route carried 7.5 million passengers in 2025.

The port is now enlarging West Harbour by 17 hectares, with two passenger terminals and four stern-ramp berths, and ending ro-pax traffic at South Harbour. The new Terminal 1 comes with 37,000 m² of building rights and 600 parking

spaces. A harbour tunnel of about €300 million will take lorries and cars straight to the main road; the Port of Helsinki bears the programme's costs. A 2021 city decision had put the tunnel at €180 million.

Other gates followed the same order. San Francisco's Ferry Building, once the second-busiest transit terminal in the world, was restored after the Embarcadero freeway came down in 1992; a US\$98 million restoration reopened it in 2003 with about 50 restaurants and food shops. In the region, Aqaba moved its port south in 2006 and sold the city-centre land, Oman decided in 2014 to move cargo from Muscat to Sohar, and Dubai ended cargo at Mina Rashid in 2018.

37,000 m²

building rights attached to Helsinki's new West Harbour Terminal 1, with 600 parking spaces

100+ ha

Jätkäsaari district on the former cargo port, planned for 17,000 to 18,000 residents

AUD 59m

cost of Sydney's Barangaroo ferry wharf, the public transport link of a 22-hectare precinct

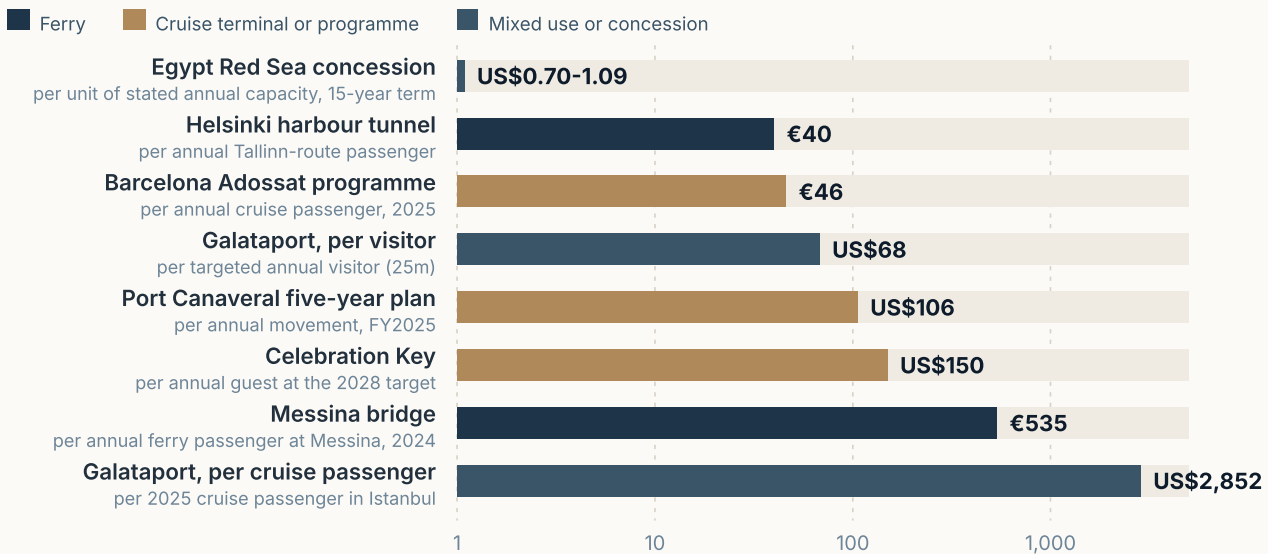
CAPITAL AND LAND

Capital per annual passenger runs from about €40 for a ferry tunnel to almost US\$3,000 for a cruise-anchored waterfront

EXHIBIT 12

Ferry infrastructure costs tens of euros per annual passenger; a cruise-led waterfront can cost thousands of dollars

Published capital cost divided by annual passengers or capacity served, log scale; currencies not converted

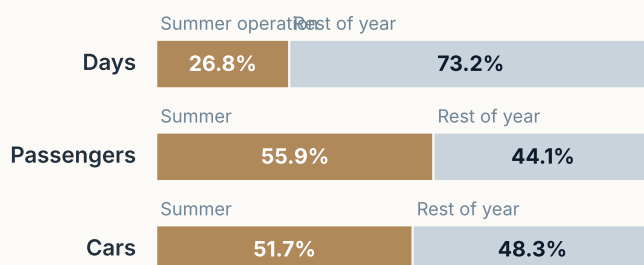


Source: Heirstone analysis of: AD Ports Group (US\$4.7m at signing, US\$3m in a Jan 2024 report; 4.3m stated capacity); Port of Helsinki (€300m; 7.5m); Port de Barcelona (€185m; 3.99m per Rispote Turismo); Galataport Istanbul (US\$1.7bn; 25m target); Canaveral Port Authority (US\$912m; 8.6m); CruiseMapper (Celebration Key US\$600m investment, first phase opened 2025; 4m by 2028); Wikipedia and Eurostat (Messina €6.1bn; 11.4m); Rispote Turismo (Istanbul 596,000). Capital is for the whole scheme, not only the terminal; a guide to scale only.

EXHIBIT 13

Over half of Tanger Med's ferry passengers arrive in a quarter of the year

Share of 2025 days, passengers and cars in the summer operation (10 June to 15 September)



Source: Tanger Med Port Authority, 2025 activity (3,220,422 passengers; 895,341 cars); Hespress FR on Operation Marhaba 2025 (Tanger Med 1.8m passengers, 463,000 vehicles). Heirstone analysis. Sources count on comparable but not identical bases.

Exhibit 12 divides capital by the annual passengers each scheme serves. Ferry figures are small because ferry traffic is large and continuous: Helsinki's €300 million tunnel works out at about €40 for each annual passenger on the Tallinn route. Cruise programmes cost more per head. Galataport works out at about US\$2,850 per 2025 cruise passenger but US\$68 per targeted visitor; on those figures its case rests on the property.

Seasonality changes the land arithmetic for ferries. Tanger Med's passenger port has eight berths on 35 hectares shared with lorry traffic, and in 2025 handled 3.22 million passengers, 895,341 cars and 535,203 TIR lorries: about 92,000 passengers per hectare. More than half of its passengers and cars came in the 14 weeks of the summer operation, so marshalling land sized for the August peak is underused for much of the year. Red Sea ferry gates serving pilgrims and seasonal workers are likely to peak in the same way, and their land plans should give the quiet months other uses.

CHAPTER 5

05

Designing in the limits

Europe's heritage ports are now paying to reduce the cruise traffic they once competed for. New gates can set their limits before the first call.

Barcelona's port and city council agreed in July 2025 to cut the cruise terminals on the Adossat quay from seven to five by 2030, demolishing three and building one public terminal for 7,000 passengers at a time. Maximum daily capacity falls 16%, from 37,000 to 31,000 passengers, in a programme carrying €185 million of public and private investment.

A cap of 31,000 binds only on peak days. Barcelona's 3.99 million cruise passengers in 2025 averaged about 10,900 a day, around a third of the cap. Cruise passengers are about 2.5% of the city's daily visitors off-season and 7.5% at peak, according to CruiseMapper, so the measures are aimed mainly at peak-day crowding.

Venice banned ships over 25,000 gross tonnes from St Mark's Basin and the Giudecca Canal in August 2021 and budgeted €157 million for temporary berths at Marghera. Amsterdam will cap sea cruise calls at 100 a year from 2026, down from 190, and move its terminal out of the centre by 2035.

EXHIBIT 14

European ports are paying hundreds of millions of euros to retrofit limits

Caps, bans and fees on cruise traffic in selected destinations

DESTINATION	MEASURE	IN FORCE	BEFORE	AFTER	PUBLIC COST OR REVENUE
Venice	Ships over 25,000 GT, 180 m or 35 m air draft barred from St Mark's Basin and Giudecca Canal	Aug 2021	n/p	Ban	€157m budgeted for up to five temporary berths at Marghera
Barcelona	Adossat cruise terminals cut from seven to five; one new public terminal	2026-30	37,000 a day	31,000 a day	€185m public-private, after €265m in 2018
Amsterdam	Sea cruise calls capped; shore power required; terminal to leave the centre	2026; 2027; 2035	190 calls	100 calls	€46-103m revenue at risk (TravelPulse)
Amsterdam	River cruise calls halved	2026	c.2,300 calls	1,150 calls	n/p
Dubrovnik	Daily ship limit	2019	n/p	2 ships a day; 5,000 to 8,000 passengers (sources differ)	n/p
Palma	Daily ship limit	2025	n/p	3 ships, only one over 5,000 passengers	n/p
Cannes	Daily cap on passenger visits	2026	n/p	6,000 a day	n/p
Bar Harbor, US	Daily landing cap by referendum; limited to July and August by a court in May 2026	2025 season	c.182,000 guests (2024)	1,000 a day; c.67,000 guests (2025)	n/p (Cruise Hive for guest numbers)
Greece	Seasonal per-passenger fee	Jul 2025	None	€20 Mykonos and Santorini in peak; €5 elsewhere	Over €50m a year expected (CruiseMapper)

Source: QualityTravel.it and Seatrade Cruise News (Venice, 2021); Port de Barcelona and Catalan News (Jul 2025); Shipping Telegraph and TravelPulse (Amsterdam, 2024); The Maritime Executive (2018), Ship Technology (2019) and Euronews (Jul 2025) on Dubrovnik; Euronews (Palma, Cannes, Jul 2025); Bangor Daily News (Feb 2024), Seatrade Cruise News (2024) and Cruise Hive (May 2026) on Bar Harbor; Proto Thema (Jul 2025) and CruiseMapper (2025) on Greece. n/p: not published.

For a new gate the lesson is to set the numbers before the first call: a daily passenger ceiling tied to what the town or site it serves can absorb, and a published per-passenger levy that pays for the public cost of transit calls. Dubrovnik, a city of about 42,000 residents, received some 1.2 million visitors in 2018, about 29 per resident, according to Ship Technology. The same ratio can be calculated for the heritage towns and sites behind the region's planned gates before their cruise traffic grows.

29

visitors per resident in Dubrovnik in 2018 (1.2m visitors, 42,000 residents; Heirstone analysis of Ship Technology figures)

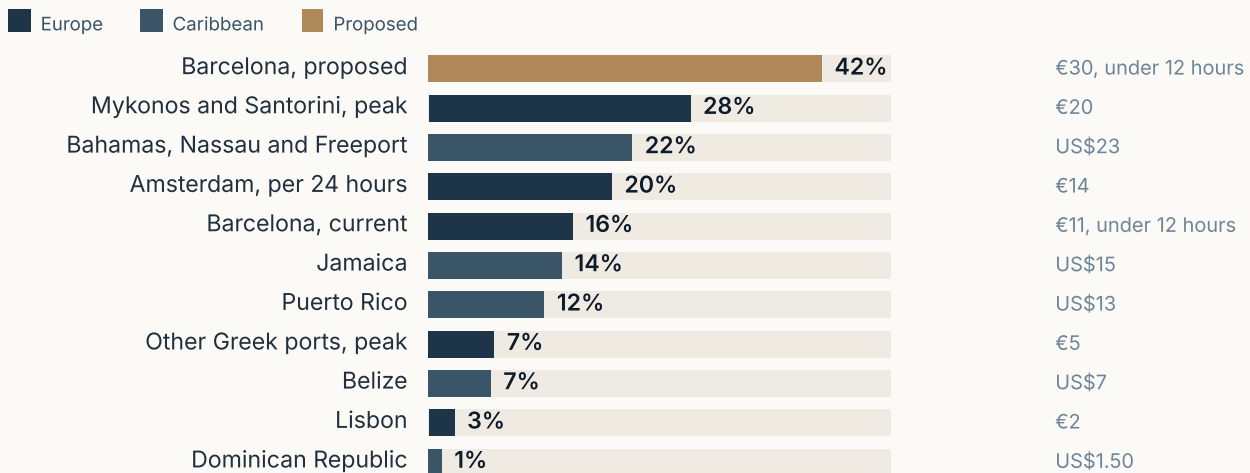
LEVIES AND CAPS

Published studies find little effect from levies on volumes; Bar Harbor's hard cap cut its cruise guests by almost two thirds

EXHIBIT 15

Barcelona's proposed €30 levy would take 42% of what a transit passenger spends ashore

Per-passenger levy or head tax as a share of average transit passenger spend (Europe €70.63; Caribbean US\$104.36); levy shown on the right



Source: Heirstone analysis of: Scottish Government, Cruise ship levies: international evidence review (May 2025); Euronews (Jul 2025; Jun 2026); Proto Thema (Jul 2025); Cruise Europe/BREA and FCCA/BREA spend averages. Levies as reported in 2024 to 2026 and may have changed since.

Greece's fee, in force since July 2025, charges €20 a head at Mykonos and Santorini from June to September, €12 in the shoulder months and €4 in winter, and €5, €3 and €1 elsewhere. It is expected to raise more than €50 million a year on more than 8 million disembarkations, according to CruiseMapper, or about €6.25 a passenger. Barcelona charges €11 to passengers in port for under 12 hours and its council has approved a proposal to raise this to €30.

The evidence on volumes is thin. A 2025 review for the Scottish Government found no appreciable change in tourist numbers after the Balearic tax, and Alaska's cruise visitation kept rising after its levy; Amsterdam's calls fell from 180 in 2018 to 128 in 2019, the year its €8 tax took effect. Hard caps act directly. Bar Harbor in Maine voted

in 2022 to limit cruise landings to 1,000 people a day; according to Cruise Hive, cruise guests fell from about 182,000 in 2024 to 67,000 in 2025.

Lines are also building destinations they control. Carnival opened Celebration Key on Grand Bahama in July 2025, the first phase of a US\$600 million investment that it expects to draw 4 million visitors a year by 2028, according to CruiseMapper. With four berths it can take over 21,000 guests a day, according to Cruise Radio. Royal Caribbean's 17-acre beach club on Paradise Island includes 4 acres of Crown Land contributed as equity, and Bahamians were invited to own up to 49% of it. Cruise Saudi opened a private beach destination about 220 miles north of Jeddah for AROYA in December 2024, according to Gulf News.

€6.25

average Greek cruise fee per disembarkation implied by the expected €50m on 8m passengers (Heirstone analysis)

-63%

fall in Bar Harbor's cruise guests in the first season under its 1,000-a-day cap

4 of 17

acres of Royal Beach Club Paradise Island contributed by the state as Crown Land equity

CHAPTER 6

06

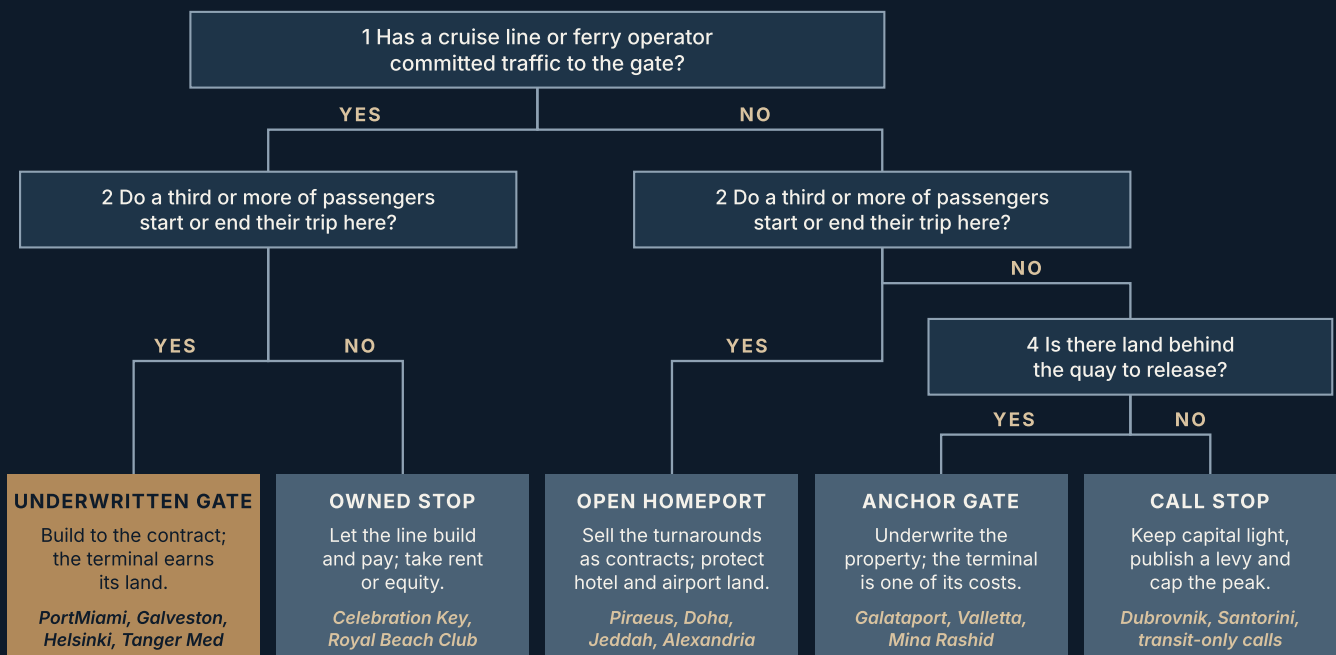
The Heirstone
Sea Gate Test

The evidence in the first five chapters reduces to five questions. Three of them sort a gate into one of five types, each with a different answer to what should be built and who should pay. The other two apply to every gate. We use the test to judge whether a proposed terminal will earn its land or become a public cost.

EXHIBIT 16

Three questions sort a sea gate into five types

Heirstone Sea Gate Test: decision path (questions 1, 2 and 4) and the action that follows for each type



Source: Heirstone analysis. Examples are classified on the public evidence in chapters 1 to 5 and the benchmark that follows. Questions 3 and 5 apply to every type.

1 Committed traffic

A cruise line guarantees passenger moves or funds the terminal, or a scheduled ferry service carries at least a million passengers a year.
Royal Caribbean 2.1m a year at PortMiami; Norwegian 1.3m to 1.8m; Tanger Med 3.22m.

2 Turnaround share

At least a third of passengers begin or end their trip at the gate. Ferry gates pass by definition.
Italy 33%; MedCruise ports 27%; Barcelona and Piraeus 58 to 59%.

3 Capture per move

The port or city collects about US\$10 or more per passenger move through a published tariff, rent or levy.
US\$14 to US\$36 in Miami; Barcelona €11; Greek average about €6.25.

4 Land in play

Developable land or building rights adjoin the terminal under the same plan and the same owner or concession.
Helsinki Terminal 1, 37,000 m²; Galataport, 400,000 m².

5 Load limits set

Daily caps and a levy are published. A new gate writes them in, with a visitor-per-resident check, before it opens.
Barcelona's €185m retrofit; Dubrovnik's two ships a day.

Thresholds are set from the evidence in this report. A gate that fails question 1 is not unbuildable: it needs a different owner, a lighter terminal or a land plan that pays for it.

THE BENCHMARK

The Heirstone Sea Gate Benchmark

Thirty-one cruise and ferry gates and Egypt's national total. Passengers as published by the port or authority, with year; bases differ (movements, visitors, customs-processed). Per call is passengers divided by calls (Heirstone analysis). Turnaround is the share of passengers or calls that start or end a trip at the gate. n/p: not published in the sources reviewed. n/a: not applicable.

EXHIBIT 17

Heirstone Sea Gate Benchmark 2026: global reference gates

GATE	COUNTRY	PASSENGERS	CALLS	PER CALL	TURNAROUND	FLAGSHIP CAPEX AND FUNDER	TERM	HEIRSTONE METRIC OR NOTE
CRUISE GATES: NORTH AMERICA, EUROPE AND ASIA								
PortMiami	US	8.56m (FY25)	n/p	n/p	Homeport hub	MSC AA c.US\$300m (2025), line	20 to 62 yrs	c.US\$36 per guaranteed move (RC)
Port Canaveral	US	8.60m (FY25)	1,038	8,287	18 homeported ships (FY26 plan)	US\$912m 5-yr plan, port	n/a	US\$21 cruise revenue per move
Galveston	US	>3.4m (2024)	380+ sailings	c.8,900	Homeport base	T16 US\$156m (2025), port on line contracts	NCL 10 yrs + options	US\$28,400 per passenger of design ship
Southampton	UK	1.9m (2019); >3.0m (2025)	c.520 (2019)	3,654	n/p	Horizon £55m (2021), ABP with lines	n/p	Sized for 4m a year within 30 years
Barcelona	Spain	3.99m (2025)	791 (2024)	4,622	58%	Adossat €185m (2026-30), public-private	n/p	€46 per annual passenger; cap 31,000 a day
Civitavecchia	Italy	3.5m (2025)	n/p	n/p	n/p	n/p	n/p	Rome's cruise gate
Piraeus	Greece	1.86m (2025)	863	2,155	59%	n/p; COSCO holds 67% of the port authority	n/p	Terminal B 1,500 passengers an hour
Istanbul, Galataport	Türkiye	596,000 (2025)	n/p	n/p	n/p	US\$1.7bn incl. US\$702m bid, private	30 yrs	US\$2,852 per cruise passenger; US\$68 per target visitor
Valletta	Malta	>0.5m a year	n/p	n/p	n/p	n/p; GPH majority since 2015	n/p	19 restored warehouses and bastions
Venice	Italy	547,587 (2024)	n/p	n/p	n/p	€157m temporary berths (2021), state	n/a	Ships over 25,000 GT barred from centre
Singapore, Marina Bay	Singapore	1.85m (2024, port)	n/p	n/p	n/p	S\$500m (2012), Tourism Board	n/p	S\$73,500 per design passenger
FERRY GATES								
Messina	Italy	11.4m (2024)	n/p	n/a	All	Bridge €6.1bn approved 2025, state	n/a	€535 per annual passenger (bridge)
Dover	UK	10.86m (2019)	n/p	n/a	All	Western Docks £250m	n/a	2.6m lorries, 2.2m cars (2017)
Helsinki	Finland	9.5m (2025)	n/p	n/a	All	Tunnel c.€300m, port	n/a	€40 per Tallinn-route passenger (tunnel)
Tallinn	Estonia	8.3m (2024)	n/p	n/a	All	n/p; state holds 67%	n/a	22% below 2019 peak of 10.6m
Tanger Med	Morocco	3.22m (2025)	n/p	n/a	All	n/p	n/a	92,000 passengers per hectare a year

Source: Heirstone Sea Gate Benchmark 2026; sources as listed for Exhibits 1 to 15 and in the notes. Venice and Singapore passengers from Wikipedia's compiled list (citing MedCruise and Singapore statistics). Metrics in the final column are Heirstone analysis unless attributed.

THE BENCHMARK, CONTINUED

Middle East and African gates: plenty of stated capacity, little published traffic

GATE	COUNTRY	PASSENGERS	CALLS	PER CALL	TURNAROUND	FLAGSHIP CAPEX AND FUNDER	TERM	HEIRSTONE METRIC OR NOTE
GULF AND ARABIAN SEA								
Dubai, two terminals	UAE	281,775 (Jan-Nov 24)	187	1,507	n/p	Mina Rashid terminal (2014)	n/p	7 ships at once; 14,000 a day design (Cruise Hive)
Abu Dhabi, two terminals	UAE	c.700,000 (2024-25)	174 plan	n/p	14% (2022-23)	Terminal opened 2015; AD Ports	n/p	Calls +27% planned, 2025-26
Doha	Qatar	>360,000 (2024/25)	87	4,138	56 of 87 calls	n/p	n/p	Old Doha Port, 1.5 km ²
Muscat	Oman	n/p	n/p	n/p	n/p	Cargo to Sohar (2014 decision)	n/a	Port converted to cruise use
RED SEA AND GULF OF AQABA								
Jeddah	Saudi Arabia	n/p (AROYA line >140,000)	n/p	n/p	AROYA base	Terminal 2021; Mawani	n/p	Rated for over 2,500 passengers
Yanbu, Dammam	Saudi Arabia	n/p	n/p	n/p	n/p	Yanbu 4 berths; Dammam 2nd terminal (plans)	n/p	Operating Cruise Saudi ports
Aqaba	Jordan	Ferry c.1.3m	n/p	n/a	All	Port moved 2006; Al Maabar	30 yrs	City-centre port land sold
Safaga	Egypt	n/p; 1.3m stated	n/p	n/p	n/p	AD Ports; US\$3m to 4.7m for 3 terminals	15 yrs	2,000 at a time; NEOM ferry 2026
Hurghada	Egypt	n/p; c.1.5m stated	n/p	n/p	n/p	AD Ports (as above)	15 yrs	City of 214,247 residents (2023)
Sharm El Sheikh	Egypt	n/p; 1.5m stated	n/p	n/p	n/p	AD Ports (as above)	15 yrs	US\$0.70-1.09 per unit of capacity (3 terminals)
Ain Sokhna	Egypt	n/p	n/p	n/p	n/p	AD Ports; US\$5.3m reported	30 yrs	Cruise and ro-ro concessions
MEDITERRANEAN AND ATLANTIC AFRICA								
Egypt, six cruise ports	Egypt	198,124 (2023)	206	962	n/p	n/a	n/a	About 5% of Barcelona (2025)
Alexandria	Egypt	>82,000 (FY25/26)	+54%	n/p	AROYA 2026: 8 of 10 calls	Port Authority	n/a	Passengers +70% on the year
Tanger Ville	Morocco	Ferry 751,000 (summer)	n/p	n/a	All	n/p	n/p	Tangier city port (2025)
Cape Town	South Africa	n/p	n/p	n/p	n/p	n/p; Transnet	n/p	Terminal beside V&A Waterfront

Source: Heirstone Sea Gate Benchmark 2026: Dubai Media Office (Nov 2023, Dec 2024); Cruise Hive (Dec 2024); CruiseMapper citing DCT Abu Dhabi (Jan 2026); Alethiad/WAM (Nov 2023); Cruise Arabia & Africa (May 2025, Dec 2025, Jun 2026); Wikipedia (Port Sultan Qaboos, Doha Port, Port of Aqaba, Hurghada, Port of Cape Town); Ship Technology (Jul 2021); The Maritime Executive (Nov 2021, Jan 2024); Arabian Business (2024); AD Ports Group and Abu Dhabi Media Office (Jun 2024); Ahram Online (Jun 2024); Seatrade Cruise News (Nov 2024); Egypt SIS via Maritimafrika (Jul 2026); Hespess FR (Oct 2025).

What the regional rows show

Few regional rows carry a full-year passenger figure for the gate itself; several are partial years, line totals or visitor counts. None carries a published line guarantee or per-passenger levy, and only the Egyptian concessions give a capital figure. The three Egyptian Red Sea terminals state 4.3 million passengers a year of capacity, 22 times Egypt's cruise traffic in 2023.

What would improve it

Port-level season totals split between transit and turnaround passengers, published tariffs and levies, and concession terms. These are routine in the European and US ports in this benchmark, and would let investors price the region's gates.

SCORING THE GATES

US homeports pass on commitment; in the region only Mina Rashid fully passes a test

EXHIBIT 18

Sea Gate Test scorecard: 16 gates

Heirstone assessment of each gate against the five tests, from public sources at October 2026

● Passes on public evidence ● Partly — Below the threshold ○ No public evidence found

GATE	1 COMMIT	2 TURN	3 CAPTURE	4 LAND	5 LOAD	TYPE ON PUBLIC EVIDENCE	MAIN EVIDENCE
PortMiami	●	●	●	●	○	Underwritten gate	Line guarantees and line-funded terminals; US\$14-36 a move
Galveston	●	●	○	○	○	Underwritten gate	Terminal 16 funded on MSC and NCLH contracts
Southampton	●	●	○	○	○	Underwritten gate	Horizon built with MSC and NCLH; terms not public
Barcelona	●	●	●	●	●	Open homeport	58% homeport; €11 levy; cap retrofitted
Piraeus	○	●	●	○	●	Open homeport	59% homeport; €5 Greek fee in peak months
Helsinki, West Harbour	●	●	○	●	●	Underwritten gate	7.5m on the Tallinn shuttle; 37,000 m ² of rights
Tanger Med	●	●	○	○	○	Underwritten gate	3.22m ferry passengers, 56% in 14 weeks
Istanbul, Galataport	○	○	○	●	○	Anchor gate	Cruise 6% of target visitors; 52,000 m ² of retail
Valletta	○	○	○	●	○	Anchor gate	19 restored waterfront warehouses
Dubrovnik	○	○	○	○	●	Call stop	Two ships a day since 2019
Dubai, Mina Rashid	○	○	○	●	○	Anchor gate	Cargo out in 2018; seven ships at once
Abu Dhabi	●	—	○	●	○	Call stop, converting	14% of 2022-23 visitors embarked; AIDA base
Doha	○	●	○	●	○	Open homeport, emerging	56 of 87 calls with a turnaround, 2024/25
Jeddah	●	●	○	○	○	Open homeport, emerging	AROYA base; turnaround share not published
Alexandria	●	●	○	○	○	Open homeport, emerging	AROYA 2026: 8 turnarounds, 2 homeported
Safaga, Hurghada, Sharm	○	○	○	○	○	Call stop	4.3m stated capacity; no traffic published

Source: Heirstone analysis using the Heirstone Sea Gate Benchmark 2026 and the sources for Exhibits 1 to 17. An open ring means we found no public evidence, which is not the same as a failure. Types follow the decision path in Exhibit 16.

PortMiami and Galveston pass the first two tests, and Miami publishes enough to price a passenger move. Helsinki and Tanger Med pass the first through their ferry timetables. Barcelona alone passes four tests, load by way of a €185 million retrofit. Galataport and Valletta pass only on land, consistent with the anchor gate type.

Most regional gates show open rings on commitment, capture and load. Alexandria and Jeddah score partly through AROYA's programme, and Doha already exchanges passengers on most of its calls. None of the Egyptian, Saudi or Emirati gates we examined publishes a line guarantee or a per-passenger cruise levy, the two items an investor would ask for first.

0 of 5

Egyptian, Saudi and Emirati rows in the scorecard with a published line guarantee or per-passenger cruise levy

1 of 16

gates passes four tests: Barcelona, its load limits by retrofit

CHAPTER 7

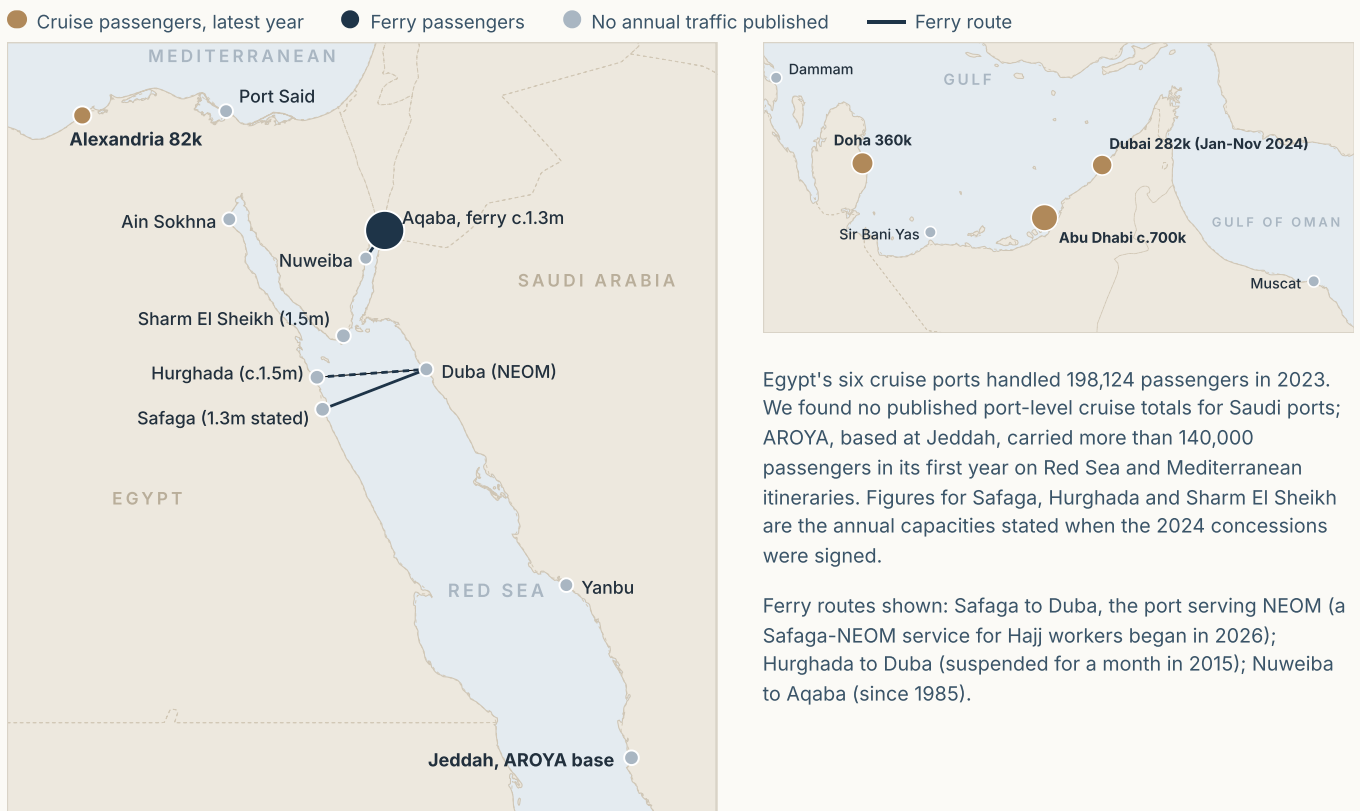
07 Egypt, Saudi Arabia and the Gulf

The region has a winter cruise season and heritage sites within reach of its ports, and public investment in terminals is under way. It has less of the contracted traffic and published data that made the Florida and Mediterranean gates bankable.

EXHIBIT 19

The Red Sea and Gulf gates: traffic is published for a few, stated capacity for several

Selected cruise and ferry gates; circle area scales with latest published annual passengers



Source: Heirstone Sea Gate Benchmark 2026; Seatrade Cruise News quoting Egyptian Ports (Nov 2024); Egypt SIS via Maritimafrica (Jul 2026); Ahram Online (Jun 2024); Cruise Arabia & Africa (Dec 2025; May 2025); Dubai Media Office (Dec 2024); CruiseMapper citing DCT Abu Dhabi (Jan 2026); Wikipedia (Port of Aqaba, Duba, Nuweiba); Baird Maritime (2009); Egypt Independent (2015); AD Ports Group (May 2026). Positions are approximate, at port level; routes are schematic. Natural Earth base.

The region's cruise traffic is concentrated in the UAE and Qatar. Abu Dhabi, Dubai and Doha together reported more than 1.3 million cruise passengers or visitors in their latest published periods, counted on different bases. Saudi Arabia is building from a low base: Cruise Saudi has facilitated nearly half a million cruise visitors from 16 lines since 2021, according to Cruise Arabia & Africa. Egypt's six cruise ports handled 198,124 in 2023.

Ferries are the other half of the regional picture and the less documented one. The Aqaba ferry terminal handles about 1.3 million passengers a year, according to Wikipedia, and Morocco's ports handled 3.2 million ferry passengers in the 14 weeks of their 2025 summer operation. Neither figure appears in cruise statistics, yet both describe sea gates with land needs of their own.

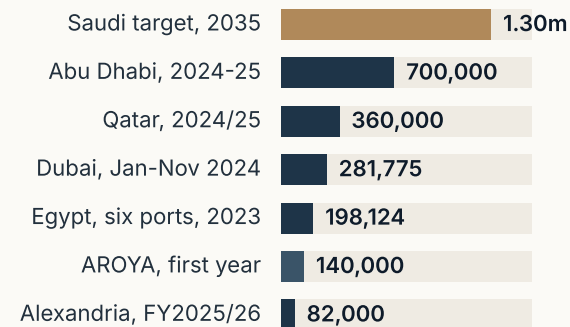
SAUDI ARABIA AND THE GULF

Cruise Saudi's target of 1.3 million cruise passengers a year by 2035 is about nine times AROYA's first year

EXHIBIT 20

The Saudi target is nearly twice Abu Dhabi's current cruise traffic

Annual cruise passengers or visitors, latest published, and the Saudi target



Source: Gulf News (2024) and Cruise Arabia & Africa (Apr 2025) for the 2035 target; Cruise Arabia (Jul 2025) cites 1.33m by 2037; CruiseMapper citing DCT Abu Dhabi (Jan 2026); Cruise Arabia & Africa (May 2025; Dec 2025); Dubai Media Office (Dubai Customs, Dec 2024); Seatrade Cruise News (Nov 2024); Egypt SIS via Maritimafrica (Jul 2026). Bases differ: arrivals, visitors, customs-processed passengers and line passengers.

9.3x

the 1.3m target relative to AROYA's 140,000 first-year passengers (Heirstone analysis)

75%

of Doha's 73 scheduled calls in 2025/26 involve a full or partial turnaround, according to Cruise Arabia & Africa

4.0x

Abu Dhabi's cruise visitors in the first quarter of 2023 against the same quarter of 2022, a restart year: 363,494 against 89,953 (Heirstone analysis of WAM figures)

10

cruise destinations planned by Cruise Saudi: three operating, four in development and three in the longer pipeline

Cruise Saudi targets 1.3 million cruise passengers a year by 2035, according to Gulf News and Cruise Arabia & Africa; a later report gives 1.33 million by 2037. It plans ten destinations: Jeddah, Yanbu and Dammam operate, four are in development and three are in the longer pipeline. Saudi Arabia's first cruise terminal opened at Jeddah in July 2021, rated for more than 2,500 passengers.

AROYA, the Saudi line launched by Cruise Saudi, began Red Sea sailings in December 2024 with a 335-metre ship that cost US\$355 million and carries more than 3,362 passengers. According to Cruise Arabia & Africa, fares start at SAR 999 for three nights, and the line carried more than 92,000 passengers in its first four months, 38,000 of them in Ramadan, and more than 140,000 in its first year. Reaching 1.3 million a year would take about nine times that volume, from AROYA and from foreign lines basing ships in the kingdom.

AROYA's passengers begin and end their cruises at the line's home ports, the higher-spending kind of traffic described in chapter 2. The commitment that would make new Saudi terminals bankable on the Florida model is the next step: guarantees or terminal funding from foreign lines that deploy ships there for the winter.

The UAE reports the region's largest published cruise volumes. Abu Dhabi received around 700,000 cruise arrivals in 2024-25 and expects 174 calls across its two terminals in 2025-26, 27% more, according to CruiseMapper citing the emirate's tourism department. Dubai Customs processed 281,775 cruise passengers and 187 ships at Mina Rashid and Dubai Harbour between January and November 2024; Mina Rashid can berth seven ships at once. Qatar drew more than 360,000 cruise visitors on 87 calls in 2024/25.

These volumes come in a winter season. A terminal sized on annual capacity sits idle for months; one sized on calls per week at the winter peak, with its hall planned for events or other uses in summer, ties up less capital.

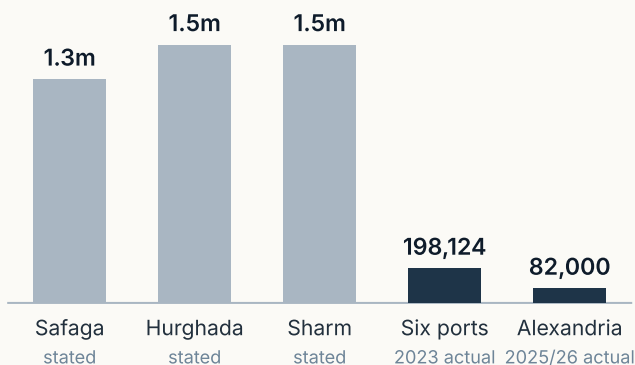
EGYPT

Egypt states capacity for 4.3 million cruise passengers a year at three Red Sea terminals; its six cruise ports handled 198,124 in 2023

EXHIBIT 21

Stated capacity is 22 times Egypt's 2023 cruise traffic

Annual cruise passengers: capacity stated at the 2024 concessions, and published traffic



Source: Ahram Online (Jun 2024); Seatrade Cruise News quoting Egyptian Ports (Nov 2024); Egypt SIS via Maritimafrika (Jul 2026). Heirstone analysis: $(1.3m + 1.5m + 1.5m) / 198,124 = 22$.

On the Red Sea, AD Ports Group took 15-year concessions in June 2024 for the cruise terminals at Safaga, Hurghada and Sharm El Sheikh, and two 30-year concessions at Ain Sokhna for ro-ro and cruise terminals. Stated capacities are 1.3 million passengers a year at Safaga and about 1.5 million each at Hurghada and Sharm El Sheikh; Safaga's hall takes 2,000 passengers at a time. Egyptian officials projected about US\$500 million of direct revenue from ship fees and services at the four ports over the concession period. AD Ports launched services in May 2026, including a Safaga-NEOM ferry for Hajj workers.

Ferries are the corridor with a record of volume, though a fragile one. Two fast ferries entered Safaga-Duba service in 2009, each carrying 1,200 passengers, 120 cars and 15 lorries,

and carried more than 69,700 passengers in their first months, many of them pilgrims. The Hurghada-Duba route was suspended for a month in 2015, according to Egypt Independent. The Aqaba-Nuweiba car ferry has run since 1985 under a joint venture of Jordan, Egypt and Iraq. Little ridership data is published for these routes.

The Strait of Gibraltar shows what a managed seasonal corridor can carry: 3.2 million passengers on 29 ships and 13 lines in Morocco's 2025 summer operation, with compulsory advance booking in the August return weeks. A Red Sea ferry plan needs the same split of demand by purpose, between pilgrims, workers and tourists, before it sizes berths and marshalling land.

Egypt's six cruise ports, Alexandria, Port Said, Sharm El Sheikh, Suez, Hurghada and Safaga, received 206 calls and 198,124 passengers in 2023, according to Egyptian Ports figures reported by Seatrade Cruise News. That is about 5% of Barcelona's 2025 traffic and under 30% of Abu Dhabi's. Alexandria is growing quickly: its cruise calls rose 54% and its cruise passengers more than 70%, to over 82,000, in the 2025/26 financial year, equal to about 41% of the country's 2023 total.

AROYA scheduled ten Alexandria calls in its 2026 season, eight using the port as a turnaround and two fully homeported there, according to Cruise Arabia & Africa, which reported that 274 Egyptians embarked at the June 2026 call and described it as the resumption of turnaround operations at the port. This season that business is led by a Saudi line.

962

passengers per call in Egypt in 2023, against 2,236 across MedCruise member ports in 2025

+70%

growth in Alexandria's cruise passengers in FY2025/26, to more than 82,000

US\$0.3m

a year: average AD Ports investment across the three Red Sea terminals over the 15-year concessions (US\$0.2m on the US\$3m reported in January 2024)

RECOMMENDATIONS

Commit the traffic, then build the hall

The region's gates need contracts that commit ships to them and land plans that put value where the passengers go. Berths and halls are already being built across the region.

Governments, port and land authorities

- 1 Tender terminal rights against minimum guaranteed passenger moves, as Miami does, and size the hall to the guarantee. Stated capacity is not a basis for capital spending.
- 2 Publish season totals by port, split between transit and turnaround passengers. Europe and the US do this routinely; the region mostly does not.
- 3 Set a per-passenger levy and a daily load limit before the first call. Venice and Barcelona are retrofitting them at €157m and €185m.
- 4 Release cargo land behind city ports in the order Helsinki followed: move the cargo first, then plan the district, the terminal and road access together before releasing plots.

Investors and lenders

- 1 Lend against contracts. Ask who guarantees the passenger moves, for how long and at what price per move; about US\$14 to US\$36 a move is the Miami range.
- 2 Where there is no guarantee, size debt to the property income rather than to projected cruise volumes.
- 3 Stress-test demand at CLIA's latest outlook of about 1.5% a year after 2025 before adding any terminal-specific growth.

Master developers

- 1 Model the 94%. A terminal-led waterfront must stand on its non-cruise visitors, offices and hotels, as Galataport's own targets imply.
- 2 Put hotel rooms and fast airport access next to any gate that aims to homeport: 58% of European homeport passengers stay 2.3 nights.
- 3 Reserve ferry marshalling land that can take parking, events or logistics outside the peak: Tanger Med takes 56% of its passengers in 14 weeks.

Cruise lines and operators

- 1 Offer volume guarantees or terminal funding in exchange for preferential berths and adjoining land, on the Royal Caribbean and MSC models in Miami.
- 2 In Saudi Arabia and Egypt, structure private beach destinations as the Bahamas did: state land contributed as equity, with local co-ownership up to 49%.
- 3 Specify shore power: 61% of the CLIA fleet can connect, 72% by 2028, yet only 33 ports had an active shore-power berth in 2025.

How Heirstone works on sea gates and waterfronts

We help port and land authorities, developers and investors decide what to build at the water's edge and what it is worth: demand studies by ship size, season and source market; homeport and transit economics; highest and best use and masterplanning of terminal land; concession and partnership structuring; and feasibility testing using DCF, IRR, NPV and Monte Carlo methods.

Get in touch

info@heirstoneconsulting.com
heirstoneconsulting.com

NOTES, SOURCES AND DEFINITIONS

Notes

Definitions

Sea gate: a cruise or ferry passenger terminal and the land and access that serve it. **Passenger movement:** one passenger embarking, disembarking or visiting in transit, as counted by the port; a homeport passenger on a round trip creates two.

Homeport or turnaround call: a call at which passengers begin or end their cruise; a partial turnaround exchanges only some of them. **Transit call:** passengers visit for the day. **Ro-ro, ro-pax:** roll-on roll-off ships; ro-pax carry passengers and vehicles. **TIR lorry:** international road freight unit. **GT:** gross tonnage. **Lower berths:** beds at double occupancy. **Stated capacity:** annual capacity announced by a port or concessionaire.

Method

Figures are as published, in the currency and period stated, and are not converted. Calculations labelled Heirstone analysis use only the inputs shown. Port counts differ in basis (movements, visitors, customs-processed passengers, line passengers; calendar or financial years) and exhibits state the basis. Per-call and per-move values ignore ramp-ups, escalation and discounting. European spend rates are applied elsewhere for illustration only. Where sources conflict we show the range or use the more conservative figure. The scorecard is our assessment from public sources; an open ring means no public evidence was found. Map positions are approximate, at port level. Figures from a single trade or press source are attributed in the text.

Use of this report

This report is published for general information. It is not investment, legal or valuation advice and should not be relied on as such. It draws only on public information. © Heirstone Consulting L.L.C-FZ, 2026. Images are illustrative.

1 TWO TRADES, ONE QUAY

Eurostat, Maritime passenger statistics (2024 data); Port of Helsinki, traffic release (Jan 2026); Spain, Ministry of the Interior, OPE 2025 closing release (16 Sep 2025); Hespress FR on Operation Marhaba 2025 (8 Oct 2025); Tanger Med Port Authority, 2025 activity; Wikipedia, Port of Dover and Port of Tallinn; ABP via Ports Europe (2026); Dubai Media Office/RTA (15 Feb 2026); Haberler.com (22 Jan 2026) and BirGün (23 Sep 2021) quoting Şehir Hatları; CLIA, State of the Cruise Industry 2024 and 2025 and press release (15 Apr 2026); TravelPulse (May 2025); Seatrade Cruise News (20 Apr 2026); eGlobal Travel Media (1 Jun 2026); Wikipedia, Icon of the Seas; Gulf Business (AROYA); Ship Technology (Jul 2021); The Maritime Executive (Jan 2024, Nov 2025); Port de Barcelona (Jul 2025); Global Ports Holding.

2 WHERE THE CRUISE BEGINS

Cruise Europe fact sheet on CLIA/BREA studies (2019 basis); PortEconomics on Port de Barcelona and Turisme de Barcelona (2014); Catalan News on the University of Barcelona study (5 Nov 2025); Euronews Travel (26 Jun 2026); FCCA press release and BREA for FCCA, Vol I (Oct 2024); Seatrade Cruise News on MedCruise (14 May 2025; 29 Apr 2026); Cruise & Ferry on Risposte Turismo (3 Apr 2026); Greek Reporter (22 Jul 2026); Canaveral Port Authority (6 Nov 2025); Houston First (2025); Cruise Arabia & Africa (23 May 2025; 12 Nov 2025); Altihead/WAM (Nov 2023); Dubai Media Office (16 Dec 2024).

3 THE GUARANTEE IS THE ASSET

WorldPorts (IAPH) on the Royal Caribbean and PortMiami agreement; Miami Today on the Norwegian, MSC, Virgin and Disney agreements; Travel Agent Central (2018); Cruise & Ferry (2018; Dec 2025); Wikipedia, PortMiami and Ocean Cay; The Maritime Executive (Nov 2025); Baird Maritime (Sep 2021); Cruise Britain (Jul 2021); Global Ports Holding, results for the year to 31 March 2024; AD Ports Group and Abu Dhabi Media Office (Jun 2024); The Maritime Executive (Jan 2024).

4 WHAT THE TERMINAL ANCHORS

Galataport Istanbul, pre-opening release; Hürriyet Daily News (17 May 2013); Wikipedia, Galataport; bne IntelliNews (Dec 2024); Port of Helsinki, development programme; Wikipedia, Jätkäsaari, West Harbour, San Francisco Ferry Building, HafenCity, Barangaroo ferry wharf, Port of Aqaba, Port Sultan Qaboos and Mina Rashid; PES-Architects; Tanger Med Port Authority; Wikipedia, Tanger Med and Strait of Messina; CruiseMapper (Jul 2025).

5 DESIGNING IN THE LIMITS

Port de Barcelona and Catalan News (Jul 2025); The Maritime Executive (Jul 2025); QualityTravel.it and Seatrade Cruise News (Jul 2021); Shipping Telegraph and TravelPulse (2024); Euronews (3 Jul 2025); Bangor Daily News (29 Feb 2024); Cruise Hive (May 2026); Proto Thema (21 Jul 2025); CruiseMapper (2025, 2026); Scottish Government, Cruise ship levies: international evidence review (1 May 2025); Cruise Radio (2025); Royal Caribbean Group investor relations; Gulf News (2024).

6 AND 7 TEST, BENCHMARK AND REGION

Sources above; Arabian Business (2024); Cruise Arabia (Jul 2025); Cruise Arabia & Africa (29 Apr 2025, 17 Dec 2025, 19 Jun 2026); The Maritime Executive (Nov 2021); CruiseMapper citing DCT Abu Dhabi (22 Jan 2026); Dubai Media Office (2 Nov 2023); Seatrade Cruise News quoting Egyptian Ports (26 Nov 2024); Egypt SIS via Maritimafrika (20 Jul 2026); Ahram Online (12 Jun 2024); AD Ports Group (19 May 2026); Enterprise (20 May 2026); Baird Maritime (2 Sep 2009); Egypt Independent (30 Apr 2015); Wikipedia, Arab Bridge Maritime, Nuweiba, Duba and Hurghada, and List of busiest cruise ports by passengers; Cruise Trade News (2024); CLIA 2025 (shore power).



Heirstone Consulting is a real estate and strategic advisory firm working across the Middle East and Africa on markets, feasibility, highest and best use, destinations and development strategy for complex and specialised real estate.

heirstoneconsulting.com

info@heirstoneconsulting.com